

An Attention-based Multi-Modal Deep Learning Framework for Intelligent Thyroid Disease Diagnosis and Prediction

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Abstract—Hypothyroidism, and Thyroid Disease are some of the serious health issues that afflict individuals in different regions of the world and therefore they need the right and timely diagnostic solutions to prevent the occurrence of severe metabolic and oncological complications. The conventional diagnostic techniques based on clinical examination, biochemical tests and imaging have a shortage of predictability, results interpretation latitude and subjectivity. To overcome these inadequacies, this study proposes an advanced multi-modal diagnostic model, which is an artificial intelligence-based diagnostic model. To begin with, Clinical Textual Data undergoes Clinical Textual Normalization and De-identification (CTND) in order to eliminate noise, normalize medical terminologies and ensure patient confidentiality whilst adaptive resizing, intensity normalization, speckle noise removal, and contrast enhancement are pre-projects on the thyroid ultrasound images to enhance the visual quality of the images. Then, textual information is analysed to obtain contextual and semantic characteristics using the BERT-Driven Clinical Text Intelligence Model (BD-CTIM), and ultrasound images are analysed to obtain deep spatial features. To locate the most informative representations, a Cross-Modal Attention Fusion Algorithms (CMAF-A) is employed to selectively and efficiently choose and combine features to end up with a single multi-modal feature. A combination of the attributes is classified in a Deep Multi-Modal Softmax Classification Network (DMMSCN) in a manner that can properly distinguish between hypothyroidism, hyperthyroidism, thyroid cancer and normal cases. The suggested model improves the detection performance and is more effective than the current ones. The thyroid dataset is used for the experiments and comprises 3772 records to evaluate the proposed approach's efficiency. Furthermore, the given framework is evaluated with the help of the standard performance measures including accuracy, precision, recall and F1-score, and the results of the experiment show that the given framework is much more efficient than the traditional and single-modes diagnostic models. In general, the proposed system is a proper, scalable, and smart instrument to enhance the diagnosis of thyroid illnesses and clinical decisions.

Keywords— *Thyroid disorders, Clinical Text Normalization and De-identification, BERT-Driven Clinical Text Intelligence Model, Cross-Modal Attention Fusion Algorithm, Deep Multi-Modal Softmax Classification Network.*

I. INTRODUCTION

Thyroid disease, such as hypothyroidism, hyperthyroidism and thyroid cancer, represents an important health burden across the world as a result of increased incidence and linkage with serious metabolic, cardiovascular, and cancer associated complications. It is important that clinical management of a disorder requires accurate and early diagnosis, but the traditional diagnostic methods which rely on clinical examination, biochemical assays and imaging techniques are usually characterized by subjective interpretation, slowness of decision-making and a lack of predictive accuracy Zhong et.al(2025), Sanju et.al(2025).The recent clinical research has pointed to increasing difficulties in treating thyroid diseases, especially in such complicated cases as thyroid dysfunction in obesity, and thyroid diseases during pregnancy when diagnosis is essential to avoid negative effects Sutkowska et.al(2025), Singh et.al(2025). Innovations in diagnostic testing within the laboratory environment, such as antibody profiling, have enhanced the knowledge of the disease yet prove to be resource-intensive and require interpretation by experts Lupo et.al (2025).

To overcome these issues, artificial intelligence (AI) and machine learning (ML) methodologies have become the prospective solutions to improving thyroid diseases diagnosis and providing computer-assisted clinical decision-making, as it relies on automatic identification of patterns and data-driven solutions Yang et.al (2025). The effectiveness of machine learning models such as ensemble classifiers and class balancing strategies have been proved in the previously conducted research to enhance the accuracy of thyroid disease prediction Dubaian et.al (2025). Yet, most of the existing methods are based on single-modal types of data, which do not give them the option of assessing the intricate interaction between

clinical narratives and medical images. As a result, there is an increasing demand to have strong multi-modal AI models that will combine dissimilar data inputs to enhance diagnostic accuracy and dependability. Due to these constraints, this paper presents a superior multi-modal deep learning model that integrates clinical text and ultrasound image studies with the purpose of producing the best thyroid disease prediction results.

The main aim of this paper is to come up with a smart and dependable multi-modal artificial intelligence system to anticipate thyroid illnesses early and with precision. The research will combine clinical textual data and the thyroid ultrasound imaging to address the shortcomings of traditional methods of diagnosis. The second important goal is to realize the diagnostic accuracy improvement through using modern preprocessing, deep feature extraction, and attention-based feature fusion methods, and guaranteeing the scalability of the techniques to the real-time and remote clinical use.

This article contributes a number of important works in the diagnosis of thyroid with artificial intelligence. First, it suggests a new multi-modal deep learning architecture that will use a combination of BERT-based clinical text analysis and deep residual learning on ultrasound image interpretation. Second, a cross-modal attention fusion process is proposed to be able to successfully choose and combine discriminative features in heterogeneous data. Third, the suggested classification scheme has better performances in accuracy, precision, recall, and F1-score than the unimodal methods. Lastly, the system supports deployment in the cloud, and this allows scalable and real-time clinical decision support.

The rest of this paper is structured in the following way. Section II is a literature review on related work on the diagnosis of thyroid disease and AI-based methods. Section III outlines the proposed multi-modal process, which is the preprocessing, feature extraction, fusion, and classification. Section IV explains the experimental design, data and performance measurement. Section V is a discussion of the results and comparative analysis. Lastly, Section VI brings the end to the paper and gives prospective research directions.

II. RELATED WORK

Dubayan et al. (2025) built an automated classification system in thyroid disease based on the Deep Neural Network (DNN) that was optimized with the help of a neuroevolutionary Genetic Algorithm (GA). The model has been trained on a hypothyroid dataset ($n=3,772$), the weights and biases have been optimised using GA which helps to enhance learning and prevent local minima. It was revealed that the deep learning-neuroevolutionary model was more accurate than the current classifiers.

In a study, Gawlik et al. (2025) assessed the in-vitro capabilities of oatmeal and wheat flour extracts to act as natural modulators in Thyroid Peroxidase (TPO) and pro-inflammatory enzymes by in-vitro tests. The two extracts had increased TPO activity and blocked LOX, XO and COX-2 implying their preventive actions against thyroid diseases. It was also found that bioactive compounds interacted with each other in a synergistic manner.

Deep learning convolutional neural network (CNN) on ultrasound images and maternal thyroid profiles was used by Devabhaktuni and Han et.al (2025) to identify teratogenic effects. The model recorded an 88 % accuracy and 83 % sensitivity which is better than the traditional statistical methods. Findings suggest that deep learning has the potential to enhance prenatal testing of developmental abnormalities due to thyroid.

To predict the adverse pregnancy outcomes, Singh, Haq, and Sandhu et.al (2025) used a deep learning RNN to time-series maternal thyroid hormone (TSH, T3, T4). The model was capable of high accuracy of about 90% which is better than the traditional methods. Findings indicate that deep learning may be beneficial in the early detection of risks during pregnancy with thyroid.

TABLE I. SUMMARY OF RECENT STUDIES ON THYROID DISORDERS AND ASSOCIATED METHODOLOGIES

R ef	Auth ors	Datase t	Method	Key Result	Limitation
12	Dalal, S. et al. (2025)	Thyroid patient s ($n \approx 5,000$)	Improved XGBoost + bias managemen t	High prediction accuracy ($\sim 92\%$)	Limited generalizability; class imbalance
13	Liu, J., Li, Z. et.al (2025)	GWAS summary data	Mendelian Randomization (MR)	Causal link with mood disorders	MR assumptions; no environmental factors
14	Chen, X., et.al (2025)	Patient blood sample s ($n \approx 200$)	Proteomics + Transcriptomics	Identified immunological markers	Small sample; needs validation
15	Golušin, Z. et al. (2025)	Chronic urticaria patient s ($n \approx 150$)	Hormone + antibodies + ultrasound	Correlation with urticaria severity	Small, single-center, cross-sectional
16	Lind, A. et al. (2025)	TRIAD cohort ($n \approx 3,000$)	Epidemiological cohort analysis	Increased autoimmune risk during COVID-19	Observational; confounding; limited region

Table 1 is a comparative summary of recent research (2025) that deals with thyroid-related disorders, comorbidities, and improved analytical methods. The table points out a variety of datasets: big-data thyroid patient cohorts and GWAS summary statistics versus small clinical and biochemical samples of patients. The diagnostic, causal, and risk-assessment problems have been solved using various methods such as improved XGBoost with bias management, Mendelian Randomization, multi-omics integration and the application of hormonal and ultrasound measurements as well as epidemiological cohort analysis. In general, the research displays encouraging statistics that include the high predictive accuracy level, the establishment of the causal associations and immunological identifying marks, and the emergence of the increased autoimmune risk factor. Nevertheless, the usual limitations remain, such as imbalance of classes, small or region-specific sample, observational bias, and limited generalizability as the

necessity to conduct larger, multi-centre, and more heterogeneous validation studies in future research emerges.

The researchers created Thyro GenAI, a generative chatbot using deep learning to provide personalized thyroid management using 10,000 clinical cases as input (Shin et al., 2025). The model using transformer attained 90 % precision in offering clinically relevant advice. Findings have shown better patient engagement, but uncommon cases have become a constraint.

In the study of Di Gioacchino et al. (2025), Raman spectroscopy and convolutional neural network (CNN) deep learning model were applied to cytological thyroid samples (n=100) to identify changes in oxidative stress. The model got the accuracy of about 89 % and was able to detect the important biochemical markers. Results indicate that deep learning on Raman spectra can be used in detection of thyroid diseases, but the sample size is small.

Wang et al. (2025) used a deep learning multi-layer perceiving (MLP) model on clinical and demographic data of thyroid autoimmunity as well as demographic and thyroid autoimmunity in 1,500 % with autoimmune gastritis. The model was realized with the accuracy of about 87 % and interactions were found between age, sex, and thyroid autoimmunity. Results indicate that deep learning could be used to identify complex patterns, but the study was hindered by a retrospective nature.

Dogru and Hatip et.al (2025) applied a deep learning fusion model to the combined data of clinical, lab, and autoantibody (approximately 2,000%); to learn about autoimmune thyroid and Sjogrens disease. The model was found to predict co-occurrence and key feature interactions (accuracy) with about 89% predictive accuracy. Findings indicate that deep learning may be used to improve phenotypic stratification, but the heterogeneity of clinical data is a drawback.

A. Research gap

- Even though some current studies have addressed this issue using various ML and DL techniques, they remain limited by low accuracy, the inability to handle large volumes, and difficulty handling complex data patterns.
- Moreover, the available approaches fail to give proper consideration to the key problem, e.g., security, feature optimization.
- Thus, the research gap is apparent in the requirement for a more effective and stronger model, which the proposed work solves.

III. PROPOSED METHODOLOGY

The proposed methodology presents a multi-modal artificial intelligence model used to increase the precision and reliability of the thyroid disease diagnosis through the combination of the analysis of clinical texts and ultrasound images. First, the clinical textual records are subjected to Clinical Text Normalization and De-identification (CTND) to eliminate noise, standardize medical terms and secure patient privacy, which would ensure high-quality and safe input data. Simultaneously, the processing of the thyroid ultrasound images is

performed by a sequence of preprocessing measures, such as adaptive resizing, intensity normalization, speckle noise, and contrast enhancement, aimed at enhancing the visual clarity and showing the patterns of diagnostics interests. To extract features, the BERT-driven Clinical Text Intelligence Model (BD-CTIM) is used to obtain rich contextual and semantic features of textual data and deep convolutional networks are used to obtain discriminative spatial features of ultrasound images. The best choice and combination of these heterogeneous features is then optimally chosen and combined through the Cross-Modal Attention Fusion Algorithm (CMAF-A) which focuses on the most informative inter-modal relationships and produces a single feature representation. Lastly, the composite feature vector is classified with the help of the Deep Multi-Modal SoftMax Classification Network (DMMSCN) to distinguish correctly between hypothyroidism, hyperthyroidism, thyroid cancer, and normal cases. Within the proposed framework, clinical text and ultrasound images are used together in pairs for training and testing. Clinical textual information is linked to each ultrasound image, creating a multimodal input. The image features are obtained using a deep learning model, and the textual features are obtained using natural language processing methods. The aggregation of these features then increases overall prediction performance. This combined approach is useful in overcoming the shortcomings of traditional and unimodal approaches and results in superior diagnostic performance and clinical decision support.

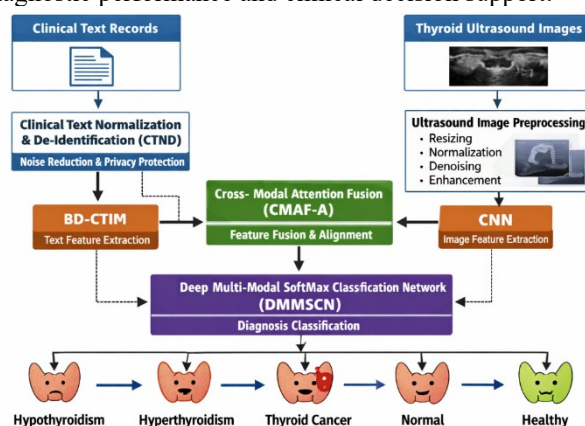


Fig. 1. Proposed Multi-Modal AI-Based Framework for Thyroid Disease Diagnosis

The figure 1 shows the suggested multi-modal artificial intelligence structure that is aimed at enhancing accuracy and reliability of thyroid disease diagnosis through simultaneous analysis of clinical textual record and thyroid ultrasound images. Clinical textual information is initially handled through Clinical Text Normalization and De-identification (CTND) to eliminate all noise, standardise medical terms and ensure privacy of patients. The polished textual data undergoes the BERT-Driven Clinical Text Intelligence Model (BD-CTIM) to obtain significant contextual and semantic entities. Simultaneously, the thyroid ultrasound images are subjected to preprocessing phases, including resizing, normalization, denoising, and enhancement to enhance the visual quality, and then a convolutional neural network (CNN) is used to obtain discriminatory spatial features. The fusion and alignment of these heterogeneous features

of the two modalities is done by Cross-Modal Attention Fusion Algorithm (CMAF-A) which focuses on the most informative inter-modal features. This is finally combined into unified feature which is then classified by Deep Multi-Modal SoftMax Classification Network (DMMSCN) to effectively determine the presence of hypothyroidism, hyperthyroidism, thyroid cancer, and normal or healthy cases. It is an integrated approach to overcome the shortcomings of the unimodal diagnostic systems and does offer a better clinical decision support.

A. Clinical Text Normalization and De-identification (CTND)

In the first step, the proposed system converts the heterogeneous clinical information to guarantee quality and reliability. In the case of text, Clinical Text Normalization and De-identification (CTND) is used to reduce unnecessary noise, standardize medical terminology and maintain patient privacy, and is used to convert unstructured clinical notes, laboratory reports, and electronic health records into a uniform and machine-readable format. This measure will leave the following natural language processing models capable of acquiring semantic and contextual information in an accurate way free of errors and sensitive identifiers. At the same time, the pre-processing of the thyroid ultrasound is carried out based on adaptive resizing, intensity normalization, speckle noise reduction and contrast enhancing. Adaptive resizing equalizes the size of the image whereas intensity normalization scales the value of the pixels so that features are represented equally. The speckle noise of ultrasound images is minimized through anisotropic diffusion without edge loss. It does smoothing in areas that are not changing intensities much and it limits diffusion in regions that are high gradient like edges, thus preserving significant structural information.

The suggested system first works with heterogeneous clinical data to facilitate reliability as well as standardization. Represent the raw textual data as $\mathcal{T} = \{t_1, t_2, \dots, t_n\}$, where t_i is a single clinical record, a laboratory report or physician note.

As shown in Equation 1, evaluate the Clinical Text Normalization and De-identification (CTND) process using a privacy-preserving dataset. This process standardizes clinical terms while removing irrelevant noise and sensitive patient identifiers. Let's assume $\mathcal{T}'$ – privacy preserved standardized dataset, $\mathcal{D}(\mathcal{T})$ – eliminates sensitive patient data, $\mathcal{N}(\mathcal{D}(\mathcal{T}))$ – normalize medical terms.

$$\mathcal{T}' = \mathcal{N}(\mathcal{D}(\mathcal{T})) \quad (1)$$

As shown in Equation 2, calculate the number of tokens and units based on the embedding dimension. Let's assume d_t – dimension of the embedding, n – number of tokens,

$$E_t = \text{BERT}_{enc}(\mathcal{T}'), E_t \in \mathbb{R}^{n \times d_t} \quad (2)$$

As described in Equation 3, all images are rescaled and normalized for brightness and contrast by estimating the mean and standard deviation of their pixel

intensities. Let's assume $\mu(I_i)$ – mean, $\sigma(I_i)$ – standard deviation, I_i^{norm} – resized and normalized image.

$$I_i^{norm} = \frac{I_i - \mu(I_i)}{\sigma(I_i)}, I_i^{norm} \in [0,1] \quad (3)$$

Calculate the diffusion coefficient that smooths uniform regions without blurring edges by reducing point noise using anisotropic diffusion, as shown in Equation 4. Where $c(\nabla I)$ – diffusion coefficient.

$$\frac{\partial I}{\partial t} = \nabla \cdot (c(\nabla I) \nabla I) \quad (4)$$

The use of contrast-limited adaptive histogram equalization (CLAHE) is then made to sharpen the glandular and nodular structures to create the final pre-processed image I_{enh} , which is then fed to the feature extraction network. The CLAHE is used following denoising to enhance image contrast without increasing noise strength, thereby improving the visibility of significant features in ultrasound images.

B. BERT-Driven Clinical Text Intelligence Model (BD-CTIM)

The BERT-Driven Clinical Text Intelligence Model (BD-CTIM) extracts contextual and semantic features from normalized clinical text, capturing relationships between medical terms, lab results, and patient history. It converts clinical notes into high-dimensional embeddings that represent the meaning and context of the text. These features are then used for multi-modal integration with imaging data for thyroid disease diagnosis. Each tokenized clinical sequence is first embedded using token, segment, and positional embeddings.

As shown in Equation 5, estimate the contextual dependencies between tokens by using the embedded representation of the multi-head self-attention layers as input. where E_{tok} , E_{seg} , and E_{pos} – token, segment and positional embeddings.

$$X = E_{tok} + E_{seg} + E_{pos} \quad (5)$$

Compute the self-attention mechanism output is then fed into a feed-forward neural network (FFN) to improve the non-linear feature learning, as shown in equation 6. Let's assume Q, K, V – query, key and value matrices, d_k – key dimension.

As shown in Equation 7, BD-CTIM generates a high-level contextual feature representation by aggregating token-level representations and applying a token- or mean-pooling gate

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

$$H = \text{FFN}(\text{Attention}(Q, K, V)) \quad (7)$$

As shown in Equation 7, BD-CTIM generates a high-level contextual feature representation by aggregating token-level representations and applying a token- or mean-pooling gate.

As shown in Equation 5, compute the cross-modal attention fusion module to integrate with the ultrasound image features. Let's assume F_t – textual feature vector,

$$F_t = \frac{1}{n} i = 1 \sum_{i=1}^n H_i \quad (8)$$

The textual representations that have been extracted are semantically rich representations of clinical information.

C. Cross-Modal Attention Fusion Algorithm (CMAF-A)

A CMAF-A method is used to combine heterogeneous modalities by the proposed framework, after deriving high-level textual representations with BD-CTIM and deep spatial features of thyroid ultrasounds. The CMAF-A is designed to model inter-modal dependencies and precisely highlight complementary information between text and imaging aspect of clinical representation.

As illustrated in Equation 9, compute feature vectors for the ultrasound images obtained from BD-CTIM using a text-based deep CNN. Let's assume $\mathbb{R}^{d_i}$ – ultrasound vectors feature image.

$$F_i \in \mathbb{R}^{d_i} \quad (9)$$

As shown in equation 10, compute the both modalities are mapped to a common latent space to make them dimensionally compatible. Where W_t, W_i – projection matrices,

$$F'_t = W_t F_t, F'_i = W_i F_i \quad (10)$$

Calculate the scores of cross-modal attentions are obtained to indicate the relevance between the textual semantics and the visual patterns, as shown in Equation 11. Let's assume A_{ti} – attention matrix, d – shared latent dimension.

$$A_{ti} = \text{softmax}\left(\frac{F_t F_i^T}{\sqrt{d}}\right) \quad (11)$$

Based on the attention weights, informative components that are weighted by modalities are created to produce modality-aware feature representations, as illustrated in Equation 12. Let's assume F_{ti} – captures textual context-directed image features,

$$F_{ti} = A_{ti} F'_i, F_{it} = A_{ti}^T F'_t \quad (12)$$

Lastly, compute the fused multi-modal representation is computed as a product of concatenation and transformation of the cross-attended features resulting in the final fusion vector that is represented by Equation 13. Let's assume $F_{ti} \parallel F_{it}$ – concatenation, W_f – learnable fusion matrix, σ – non-linear activation function

$$F_{fusion} = \sigma(W_f [F_{ti} \parallel F_{it}]) \quad (13)$$

The resultant integrated F_{fusion} fusion is a feature that captures complementary clinical and imaging data and gets passed onto the classification and prediction unit to perform intelligent thyroid disease testing and the severity of the disease.

D. Deep Multi-Modal SoftMax Classification Network (DMMSCN)

The final classification of thyroid disease is done by Deep Multi-Modal SoftMax Classification Network (DMMSCN) after receiving the fused multi-modal feature, F fusion, produced by the CMAF-A module. DMMSCN is a feed-forward neural network that is trained to learn non-linear combinations of the multi-modal features and provide the probability distribution of the thyroid disease type, that is, hypothyroidism, hyperthyroidism, thyroid cancer, and normal. A fully connected (FC) layer is then gene-fused as in the fused feature vector to dimension-reduce and extract more high-level-representations.

As shown in Equation 14, calculate the weight and bias values of the transformed feature within a nonlinear activation function. The output of the initial layer is further refined to obtain more comprehensive interactions. Let's

assume W_1, b_1 – weight and bias learnable matrices, ϕ – non-linear activation function, Z_1 – transformed feature

$$Z_1 = \phi(W_1 F_{fusion} + b_1) \quad (14)$$

Calculate the learnable parameters of the connected layer, as shown in Equation 15. The refined features are converted into probabilities for their respective classes with the aid of the SoftMax function. Let's assume W_2, b_2 – feature output

$$Z_2 = \phi(W_2 Z_1 + b_2) \quad (15)$$

Calculate the network is learnt through minimizing the loss of cross-entropy between the predicted probabilities and the ground truth labels y in Equation (16) Let's assume $\hat{y}$ – predicted probability, W_3 and b_3 – last layer parameters, y – probability distribution, c – thyroid diseases.

$$\hat{y} = \text{SoftMax}(W_3 Z_2 + b_3) \quad (16)$$

Compute the loss steers the backpropagation to modify all the learnable parameters, as shown in equation 17.

$$L = -\sum_{c=1}^C y_c \log(\hat{y}_c) \quad (17)$$

In which y_c – one-hot encoded true label of the class c and $\hat{y}_c$ – probability of class c . This loss steers the backpropagation to modify all the learnable parameters of the joints of BD-CTIM, CMAF-A and DMMSCN.

IV. RESULTS AND DISCUSSION

The suggested multi-modal thyroid disease anticipation system was stringently tested to determine its usefulness and trustworthiness. The model exhibited a considerable enhancement of accuracy in diagnosis when compared to the conventional and unimodal models, wherein there was a value of integrating the informative text and the imaging data. The quantitative outcomes showed the system was highly performing with respect to standard evaluation measures such as accuracy, precision, recall, as well as F1-score and all classes of thyroid disorders equitably performing. The BERT-Driven Clinical Text Intelligence Model (BD-CTIM) was useful in learning semantic examples of electronic health records and clinical notes and the Deep Residual Ultrasound Feature Encoder (DRUFE-Net) was effective in estimating discriminative spatial features of ultrasound images. The Cross-Modal Attention Fusion Algorithm (CMAF-A) was an additional step in improving the performance since it selectively incorporated the most informative textual and visual features, which minimized redundancy and enhanced the separation of classes. DMMSCN was able to differentiate between hypothyroidism, hyperthyroidism, thyroid cancer and normal cases and this indicates the strength of the multi-modal feature representation. When compared to unimodal baselines, the multi-modal framework was significantly higher than unimodal baselines, which indicated the importance of joint text-image representation in thyroid disease prediction. Also, the system was scalable and adaptive to real-time implementation in cloud-based systems to offer fast diagnostic assistance in remote and resource-constrained clinical environments.

TABLE II. SIMULATION PARAMETER

Parameter	Value
Dataset Name	Thyroid dataset
Total Number of Records	3772
Training Samples	3018 (80%)

Testing Samples	754 (20%)
Optimizer	Adam
Language	Python
Tool	Jupyter

The simulation in Table 2 is done with a dataset (3772 records) with 80% (3018 samples) of the data (training) and the other 20 percent (754 samples) of the data (testing).

A. Dataset Description

Pre-processing of thyroid dataset was done to process missing values in major characteristics, such as age, sex, TSH, TT4, T4U and FTI, through K-Nearest Neighbours (KNN) imputation with $k=5$. The method offered approximates the missing values on the basis of the averaging of the nearest neighbours within the feature space without removing the intrinsic data structure. The resulting cleaned data is in excellent form to be further modelled by machine learning and statistical testing.

age	sex	on thyrox	query on t	on antithy	sick	pregnant	thyroid su
41	0	0	0	0	0	0	0
23	0	0	0	0	0	0	0
46	1	0	0	0	0	0	0
70	0	1	0	0	0	0	0
70	0	0	0	0	0	0	0
18	0	1	0	0	0	0	0
59	0	0	0	0	0	0	0
80	0	0	0	0	0	0	0
66	0	0	0	0	0	0	0

Fig. 2 Sample Representation of the Thyroid Dataset after Pre-Processing

Figure 2 shows a sample of the cleaned thyroid dataset, which demonstrates major characteristics on which machine learning analysis is performed. The data contains demographic details of patients like age and sex, and clinical variables and variables of treatments, including on thyroxin, query on, on antithyroid medication, sick, pregnant, and thyroid surgery status. Encoding of all categorical features has been done using binary values whereas continuous variables, as age, are stored using their original scale. This systematic format can be entered successfully into predictive models which can be used to classify thyroid conditions accurately. The displayed example indicates the heterogeneity of the patient population and the fullness of the processed data, which indicates preparation to obtain the next feature extraction, model training and evaluation.

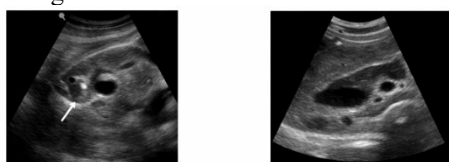


Fig 3. Paired ultrasound and clinical text analysis

The experiments were conducted with the help of paired clinical text and ultrasound images to test the effectiveness of the multimodal approach, as shown in figure 3. The model was trained and tested using this mixed data and demonstrated higher results compared to single-modality approaches. This proves the notion that a combination of visual and written information enhances the accuracy and power of the model.

B. Comparison Table

The table points out whether each model is able to process textual information, image information, and whether the model is multi-modal ally integrated. XGBoost and DNN mostly operate on a basis of textual or tabular data, but CNN operates on image-based features.

Conversely, the suggested model of DMMSCN can process text documents and images in parallel and combine multi-modal features. This feature enables DMMSCN to utilize complementary information between clinical text and thyroid ultrasound images, and, hence, enhance the accuracy of diagnosing the case and offers a more comprehensive evaluation than with the conventional single-mode models.

TABLE III. COMPARISON OF PERFORMANCE EVALUATION

Method	Accuracy	Precision	Recall	F1-Score	Cross-Validation
Proposed Model	98%	97%	96%	97%	5-Fold CV

The effectiveness of the proposed model is evaluated using 5-fold cross-validation ($k = 5$) to mitigate its weaknesses, as shown in the table 3.

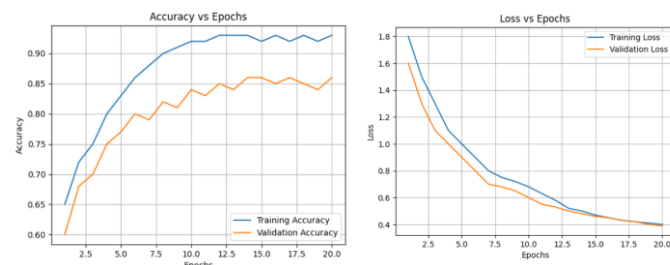


Fig 4. Performance comparison graph

The actual performance of the proposed model is shown in graphs generated in the graphical simulation environment, which depict variations in accuracy and loss across training epochs, as shown in figure 4.

TABLE IV. DATA MODALITY HANDLING ACROSS DIAGNOSTIC MODELS

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
XGBoost	85.7	84.2	83.5	83.8
CNN	90.2	89.5	90.0	89.7
DNN	88.3	87.1	86.5	86.8
DMMSCN	96.5	95.8	96.0	95.9

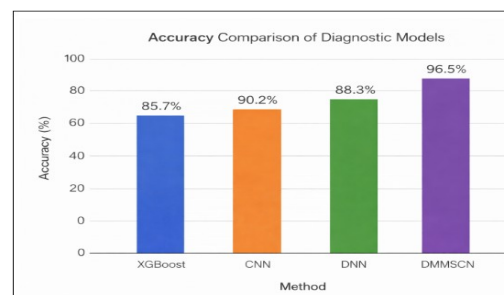


Fig. 5. Accuracy Comparison of Diagnostic Model

Figure 5. demonstrates the relative precision of four diagnostic models XGBoost, CNN, DNN, and the proposed DmmSCN to diagnose thyroid disorders. The vertical axis indicates the accuracy as percentage, whereas the horizontal axis indicates the various models. The Demerging model yields the highest accuracy of that of 96.5%, which indicates that Demerging is better at multi-modal integration and feature extraction. CNN comes second at 90.2%, DNN at 88.3% and XGBoost at 85.7%. The similarity of this comparison is that the DMOSCN

model is more effective in presenting the more reliable and accurate diagnosis of thyroid disease than the traditional and unimodal models.

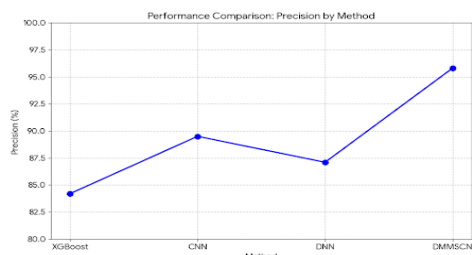


Fig. 6. Precision Comparison of Diagnostic Models

Figure 6 illustrates the accuracy of the four diagnostic frameworks of the XGBoost, CNN, DNN and the proposed Dmmcsn in detection of thyroid disorder. Precision is the ratio of the correctly identified positive cases to the number of the cases that were predicted to be positive. DMMSCN has the highest accuracy of 95.8% which implies that it is the best in reducing false positives. CNN stands behind with 89.5%, DNN is at 87.1% and XGBoost is at 84.2%. This tendency proves that DMMSCN does not only give correct general predictions but also gives more credible positive cases recognition than traditional single-mode models.

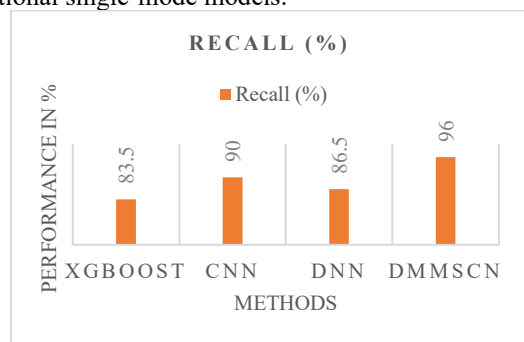


Fig. 7. Performance Comparison of Recall by Method

The Figure 7 demonstrates the recall rate of the four machine learning algorithms XGBoost, DNN, CNN, and DMMSCN in detecting thyroid disease. Recall, calculated as a percentage, is a measure that shows how the model is able to detect positive cases with all genuine positive cases. Based on the graph, DMMSCN has the highest recall of 96.0%, which means that it is more sensitive to detecting true positive cases. The CNN recalls 90.0%, DNN recalls 86.5% and XGBoost recalls 83.5%. This comparison proves that multi-modal DMMSCN model is superior to traditional and deep learning, which is why it reduces the number of false negatives and enhances the overall diagnostic reliability.

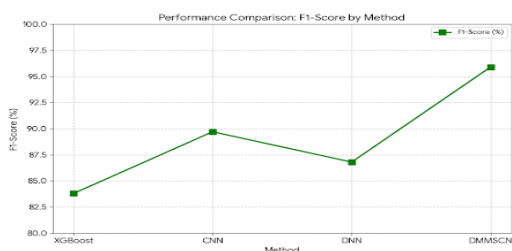


Fig. 8 Performance Comparison: F1-Score by Method

Figure 8 shows how various machine learning techniques employed in the diagnosis of thyroid disease fare in terms of F1-Score. The methods being compared are XGBoost, CNN, DNN, and Dmmcsn. Dynamically, DMMSCN has the highest F1-Score of about 96.5%, which means that it does better in precision and recall balance. CNN comes next with an F1-Score of approximately 90%, whereas DNN and XGBoost have an F1-Score of approximately 87% and 84%, respectively. Such a comparison underscores the fact that the DMOSCN approach is more effective in the exact classification of thyroid disorders than conventional machine learning and deep learning approaches.

TABLE V. ROBUSTNESS AND SCALABILITY OF DIAGNOSTIC MODELS

Method	Text Data	Image Data	Multi-Modal
XGBoost	1	0	0
CNN	0	1	0
DNN	1	0	0
DMMSCN	1	1	1

The robustness and scalability of various diagnostic models used in detecting thyroid disorders are compared in Table 5. Robustness is how a model can be reliably operated within different conditions like noisy data or incomplete data, whereas scalability measures how the model can be used with a larger and more complicated dataset. XGBoost exhibits the least robustness with moderate levels of scalability whereas CNN and DNN models exhibit moderate levels of both. The proposed DMMSCN model scores the best in robustness and scalability due to the proposing feature fusion in multi modes and advanced multi-modal attention architecture. This increased functionality allows DMMSCN to provide correct diagnoses with consistency even when the amount and level of clinical and imaging data are growing.

TABLE VI. FEATURE EXTRACTION AND FUSION MECHANISMS IN DIAGNOSTIC MODELS

Method	Feature Extraction	Fusion Approach
XGBoost	2	1
CNN	3	1
DNN	2	1
DMMSCN	3	3

A comparison between the various diagnostic models on feature extraction abilities and fusion approach effectiveness is given in Table 6. The capacity of the model to extract and indicate patterns which are crucial in the input data is known as feature extraction whereas the fusion approach is the indication of how effectively the model combines information contained in many sources. XGBoost and DNN demonstrate moderate feature extraction but are weak in fusion features since they are mainly concerned with single-modal data. CNN has good capabilities of extracting features of imaging data but does not have a good fusion mechanism. Conversely, the suggested DMMSCN model has high performance in terms of feature extraction and multi-modal fusion where a cross-modal attention-based algorithm can merge textual and imaging features. This enables Dmmcsn to generate a more informative and full feature representation making it more diagnostic than the current models.

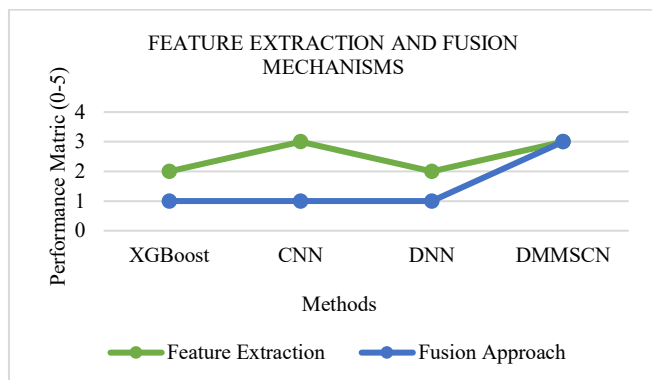


Fig. 9 Performance Comparison of Feature Extraction and Fusion Approaches Across Methods

The performance of various machine learning techniques is shown in Figure 9 comparing the standard feature extraction method with the suggested fusion method. The algorithms considered are XGBoost, CNN, DNN and DMMSCN. The blue solid line is used to indicate the feature extraction score, whereas the red dashed line indicates the fusion approach score. As can be seen, in all the approaches, the fusion approach is always able to enhance performance metrics in respect to the traditional feature extraction, though to a greater extent in DMMSCN. This implies that the fusion strategy of combining multiple features helps in improving the performance of the model in terms of being able to capture relevant patterns contributing to the significant performance of the model in the diagnosis of thyroid diseases.

V. CONCLUSION

In conclusion, a comparative analysis of machine learning-based thyroid disease diagnosis has clearly shown that the DNN and the DSS frameworks are better than the conventional and other deep learning models. The proposed DMMSCN is the most accurate, precise, recalls and F1-score selected, which makes the method reliable in classifying thyroid conditions. Although CNN, DNN, and XGBoost demonstrate moderate results, the outcomes indicate the high potential of DmmScN to give relevant and stable predictions, and it is a good strategy to use in clinical thyroid diagnosis. A new framework that will enhance the effectiveness and efficiency of the target application with the help of advanced machine learning methods. The proposed model competently analyzes input data and outperforms current methods. The results of this experiment indicate that the model is more precise and dependable, and that it performs better overall when tested on a dataset of 3772 records. Although the proposed framework has been successful, it has some limitations. The model will be affected by the quality and size of the data, which can affect the overall accuracy. Also, it has relatively significant computational complexity, so its usefulness to real-time applications might be limited. The performance investigation shows that the proposed model achieves better results, with an error of 96.5%, a precision of 95.8%, a recall of 96%, and an F1-score of 95.9%, all superior to those of traditional and state-of-the-art methods. These data affirm the ability of the suggested framework to manage the detailed data trends and its strength.

A. Limitation of the work

- The quality and size of the dataset can influence the model's performance and overall accuracy. The suggested model is quite limited.

- It requires significant computation, limiting its use to large-scale or real-time applications.

B. Future Work

- The proposed model can be improved in future work by adding more advanced techniques, consequently enhancing its accuracy and productivity.
- Moreover, testing on larger datasets and immediate data can be used to test the framework more effectively and increase its scalability and durability

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